AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 2 · STUDENT LESSON

Architecture and data flow

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Name each component. Follow each arrow. Inspect each stop.

Trace how a service keeps invalid or restricted data out of downstream work and checks answers against approved evidence.

Keep one habit: Component contract → ordered flow → complete fixtures → library picture → trace probes → local simulator → independent earliest stop.

MAIN LESSON · SECTION 1

Follow the data before trusting the answer

> Each component needs a job, an input and an observable failure path.

Learning goals

I can explain the **mechanism** and terminology of architecture and **data flow**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve service boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Requirements and service boundaries. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain architecture and **data flow** using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of architecture and **data flow**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Assign responsibilities and draw failure paths

> A trust boundary separates what different actors may access or control.

Begin with the school-office analogy

Think of a school office with a public noticeboard and a separate teacher folder. A student’s question should never send the teacher folder into a drafting system. Crossing out a citation after drafting is too late: the restricted content already crossed the boundary. Draw the route before building it, then inspect actual traces to check that each required stop happened in the right order.

Define architecture and trust boundaries

Architecture assigns responsibilities to **component**s and shows information crossing trust boundaries. A diagram should include failure paths.

Complete component contract

Component	Responsibility	Stop or alternate outcome
Request validator	Check the request shape and required fields	Invalid request stops before lookup
Role filter	Use trusted caller identity to restrict accessible source IDs before content lookup	Denied request stops before restricted source content enters context
Source lookup	Read only an approved accessible record	Missing approved record returns review
Drafting step	Phrase the approved facts; keep source content as data	Cannot invent permission or replace evidence

Component	Responsibility	Stop or alternate outcome
Response validator	Check required output fields and agreement with the supplied evidence	Mismatched draft returns review
Result	Return only an allowed checked response	Do not hide an earlier boundary violation by changing its citation

Name given and missing evidence

> **List the given facts and one fact that is not supplied.**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace a good path and the earliest point that must stop

> The caller’s access rights come from a trusted context, not a field they invent.

Original stage sequence and source-as-data rule

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Complete ordered flow and failure exits

Put the access check before the source body and drafting context

STAGE	INPUT → DECISION → OUTPUT
1. Request validator	Structured request → valid shape? → continue or invalid_request
2. Role filter	Trusted caller role + source access metadata → permitted source IDs or denied
3. Source lookup	Permitted source ID → approved body or review if absent
4. Draft	Approved facts → candidate wording/fields
5. Response validator	Candidate + approved facts → agreement? → result or review

No final-output edit can undo earlier exposure of restricted source content. Compare these arrows with recorded execution events.

All local simulation inputs

Local simulation fixture	Supplied value
Source S	student-visible, time 16:00
Source T	teacher-only, time 18:00
Trusted caller role for main probes	student, provided separately by the test harness
Request shape	Exactly one field: source_id, a nonempty string
Valid response shape	source_id and time must exactly match the accessible source
No permitted source	review, without drafting a guessed time
Scope	Local deterministic simulation; no authentication, model call or deployed service

State the contract before tracing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Architecture and data flow — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A student request bypasses the role filter and sends teacher records directly to a generator.	Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.	The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Move one arrow and predict the resulting failure

> Keep the same approved sources and trusted caller role.

Current task and response

- A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**
>
> _____
>
> _____
>
> _____
>
> _____

Predict the disconnected retrieval picture

- Change: Disconnect retrieval from the application.

▼ **Check after predicting**

Prediction: Return review status rather than inventing a time.

Reason: The supported-answer path lacks its required **evidence input**.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare an access-boundary failure with a missing-evidence failure

> Both need the earliest appropriate stop, for different reasons.

Main bypass scenario and corrected route

A student request bypasses the role filter and sends teacher records directly to a generator.

The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Keep the trace’s scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate library case and supplied schedule

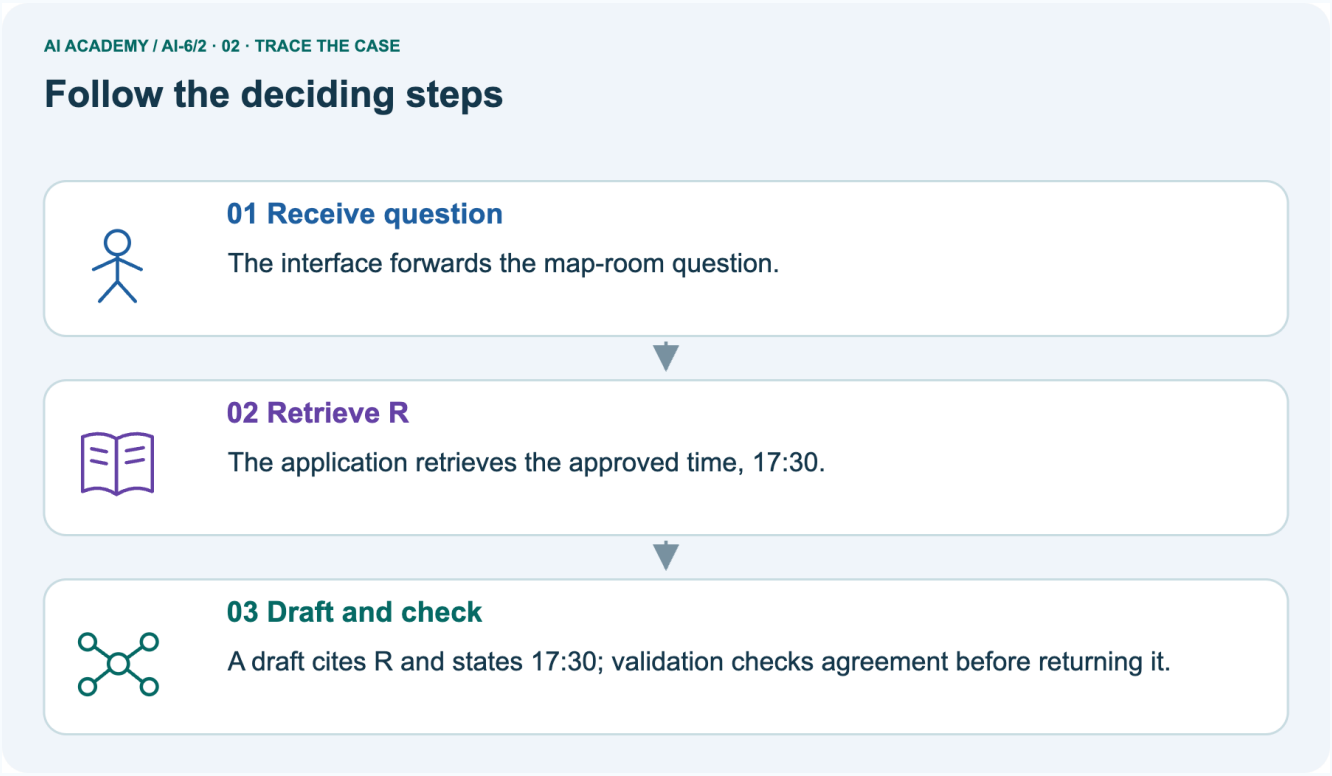
Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.

Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

The main scenario diagnoses a student path that bypassed access filtering; restricted teacher records must not reach drafting context. The picture is a separate library reply: approved schedule R says the map room closes at 17:30. Retrieval supplies that fact, drafting supplies wording, and validation checks agreement. If retrieval is disconnected, return review rather than invent a time.

The picture’s R/17:30 is separate from the simulation’s S/16:00 and T/18:00. None of these fictional records verifies a current real closing time.

Trace question, evidence, draft and check



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Receive question	The interface forwards the map-room question.
2. Retrieve R	The application retrieves the approved time, 17:30.
3. Draft and check	A draft cites R and states 17:30; validation checks agreement before returning it.

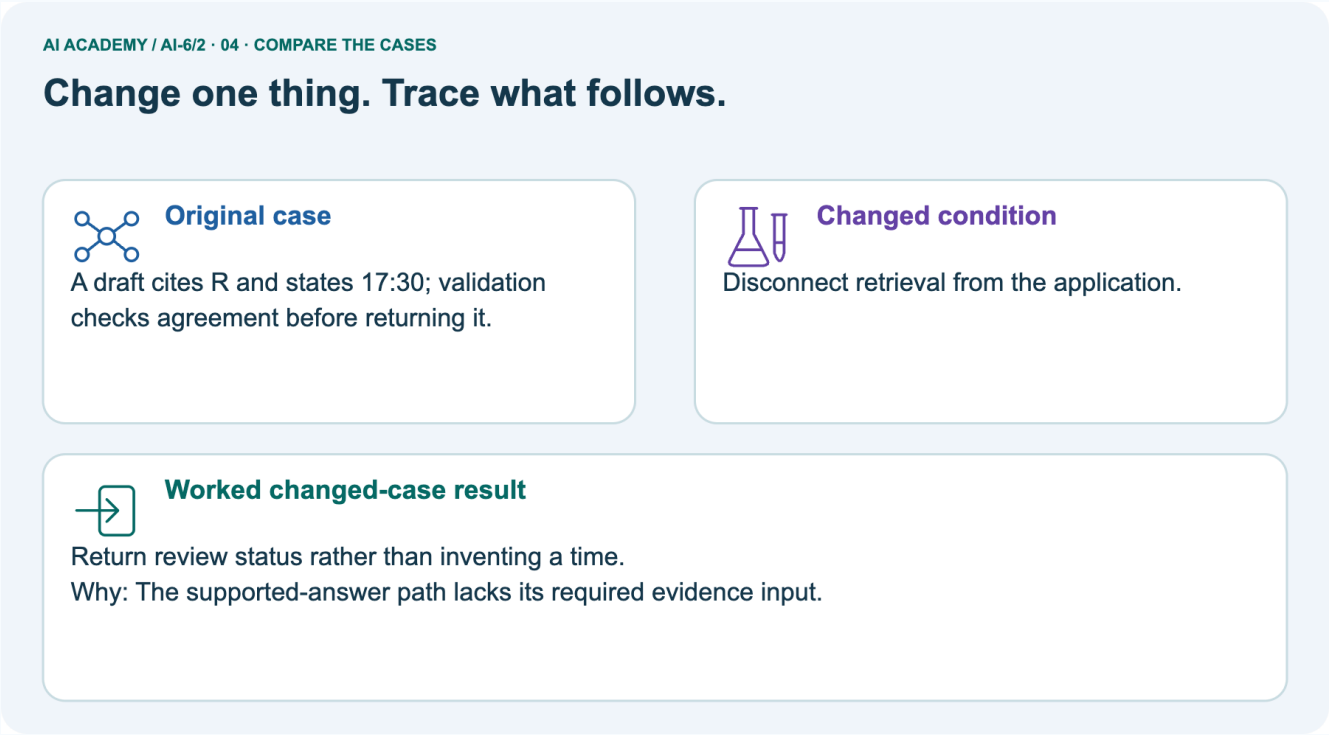
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

Predict disconnected retrieval

- Change one thing: Disconnect retrieval from the application. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what travels along each arrow and the diagram limit

- A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

- Read the labelled picture: Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

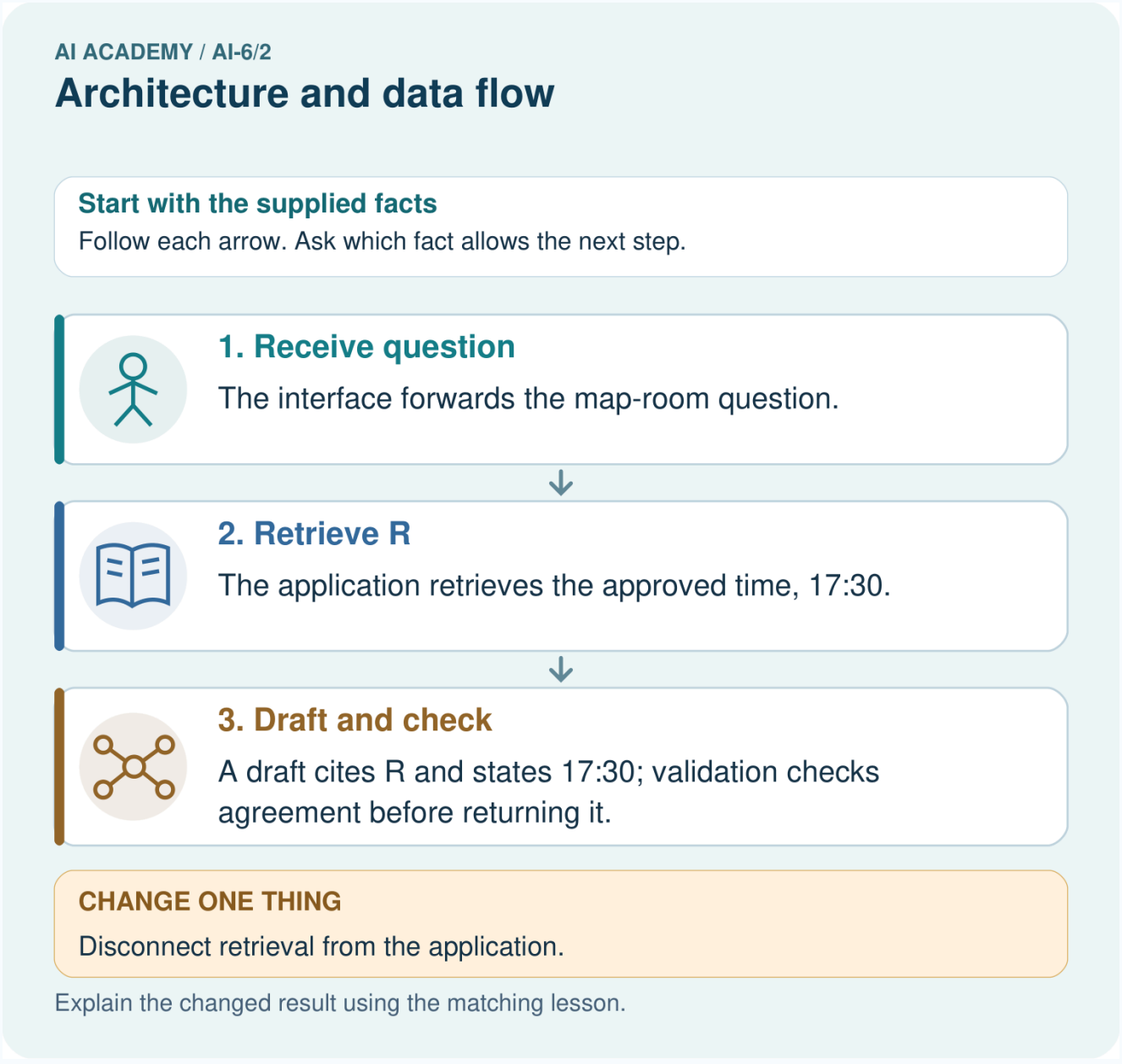
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Architecture and data flow

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Check the implementation trace against the diagram

> A component box does not prove that its check ran.

Core plan-versus-execution qualification

A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete positive, denied, missing and mismatch probes

Complete local probe	Expected stages and outcome
Student requests S	Validate → filter → lookup → draft → output check → answered S/16:00
Student requests teacher-only T	Validate → filter → denied; no source body lookup or drafting
Student requests missing X	Validate → filter → lookup → review; no drafting
S draft incorrectly says 19:00	Output check returns review, without publishing mismatched time
Request contains a caller-written role=teacher field	Invalid request shape; caller cannot promote their trusted role
Diagram claims access filtering but trace omits it	Implementation diverges from plan; preserve the failing trace and repair the route

Complete local probe	Expected stages and outcome
A source text requests new permissions	Treat it as data, not an authorized control instruction; this simple simulator is not a complete attack defense

Guided explanation and repair

- A report omits this qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise component jobs and earliest-stop reasoning

> Name what crosses an arrow and which component may authorize it.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Architecture and **data flow**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual diagram-versus-evidence check

- Use the worked case for Architecture and **data flow**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Library picture response

- Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

> **My explanation using a pictured step and the given facts**

Reserve independent malformed-input case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete local data-flow audit

> Retain both successful and stopped traces with their expected stage boundaries.

Current bypass investigation

A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

Complete runnable simulator and stage assertions

```
"""Local data-flow simulator with explicit, trusted caller-role input."""
SOURCES={'S':{'visibility':'student','time':'16:00'},
         'T':{'visibility':'teacher','time':'18:00'}}

def serve(request,caller_role,trace,*,draft=None,sources=SOURCES):
    trace.append('request_validator')
    if (not isinstance(request,dict) or set(request)!={'source_id'}
        or not isinstance(request['source_id'],str) or not request['source_id']):
        return {'status':'invalid_request'}
    trace.append('role_filter')
    if caller_role not in ('student','teacher'):
        return {'status':'denied'}
    allowed={key for key,meta in sources.items()
             if meta['visibility']=='student' or caller_role=='teacher'}
    key=request['source_id']
    if key in sources and key not in allowed:
        return {'status':'denied'}
    trace.append('source_lookup')
    if key not in allowed:
        return {'status':'review','reason':'No approved source'}
    record=sources[key]  # body read occurs only after authorization
    trace.append('draft')
    output=({'source_id':key,'time':record['time']} if draft is None else draft)
    trace.append('response_validator')
    if (not isinstance(output,dict) or set(output)!={'source_id','time'}
        or output['source_id']!=key or output['time']!=record['time']):
        return {'status':'review','reason':'Draft does not match approved evidence'}
    trace.append('result')
    return {'status':'answered',**output}

traces={}
for name,request,role,draft in [
```

```
  ('allowed',{'source_id':'S'},'student',None),
  ('restricted',{'source_id':'T'},'student',None),
  ('missing',{'source_id':'X'},'student',None),
  ('wrong_draft',{'source_id':'S'},'student',{'source_id':'S','time':'19:00'}),
  ('spoofed_request',{'source_id':'T','role':'teacher'},'student',None)]:
     trace=[]
     result=serve(request,role,trace,draft=draft)
     traces[name]=(trace,result)
     print(name,trace,result)
assert traces['allowed'][1]=={'status':'answered','source_id':'S','time':'16:00'}
assert traces['restricted']==(['request_validator','role_filter'],{'status':'denied'})
assert traces['spoofed_request']==(['request_validator'],{'status':'invalid_request'})
assert 'draft' not in traces['missing'][0]
assert traces['wrong_draft'][1]['status']=='review'
print('Simulation: role is supplied by a trusted test harness, not authenticated here.')
print('No model call, network, production access control or complete injection defense is
demonstrated.')
```

Expected traces and simulation limits

1. Predict the event sequence before running or tracing each fixture. Check that student access to T stops before source_lookup and draft; missing X stops before draft; the incorrect S time returns review at response_validator. Compare event lists with the diagram, not just final wording. The role is supplied separately by a trusted local test harness; authentication is not implemented.

   Source filtering uses access metadata before reading the selected body. Drafting is a deterministic stub and exact-field checks cover only this toy time record. These traces do not establish production access control, general semantic correctness or complete injection resistance.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: locate the earliest stop for malformed input

> Trace the new failure before opening feedback.

Complete independent malformed-request scenario

Input is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

First attempt and support record

> **My independent explanation, trace and limits**
>
> ---
>
> ---
>
> ---
>
> ---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

Input is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

> **My independent procedure, result and limitation**
>
> ---
>
> ---
>
> ---
>
> ---

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve
</details>

MAIN LESSON · SECTION 11

Save an architecture diagram with matching execution traces

> Keep component contracts, failure exits and evidence sources together.

Project task and artifact

Add a labelled architecture and **data flow evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of architecture and **data flow**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a route you can explain and inspect

> A correct diagram and a matching trace are separate artifacts.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of architecture and **data flow**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Component	A unit with a declared responsibility
Data flow	The path information takes through components
Trust boundary	A separation between differently authorised actors or data

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Architecture and data flow — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A student request bypasses the role filter and sends teacher records directly to a generator.	Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.	The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.

By the end, I can	How I will show it
Explain the mechanism and terminology of architecture and data flow	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.
- Which labelled step matches this detail? A draft cites R and states 17:30; validation checks agreement before returning it.
- 1. Retrieve R
- 2. Draft and check
- 3. Receive question
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Draft and check A draft cites R and states 17:30; validation checks agreement before returning it.

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

Changed-case question and optional hint

- Changed-case practice: Disconnect retrieval from the application. Which conclusion is supported by the supplied facts?
- 1. Return review status rather than inventing a time.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return review status rather than inventing a time. The supported-answer path lacks its required **evidence input**.

Repair and retain the record

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

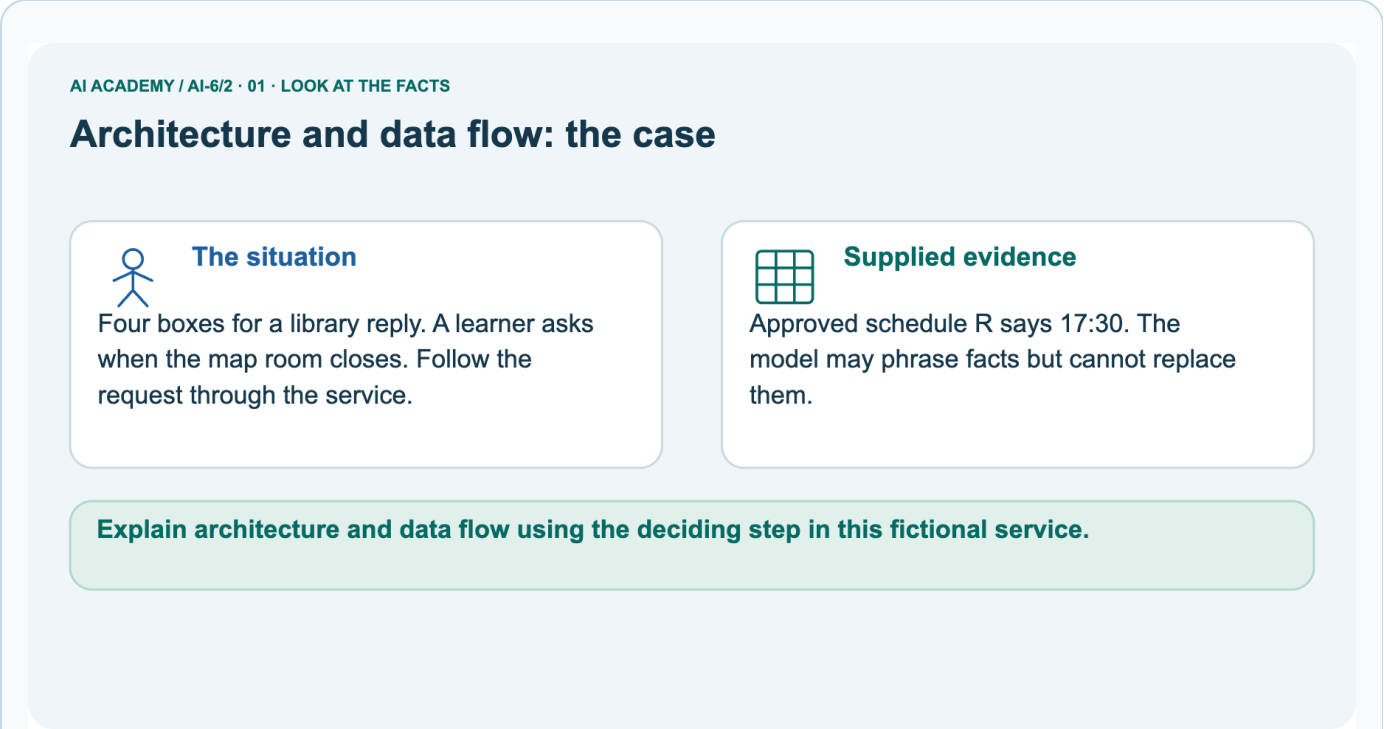
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Architecture and data flow: the case

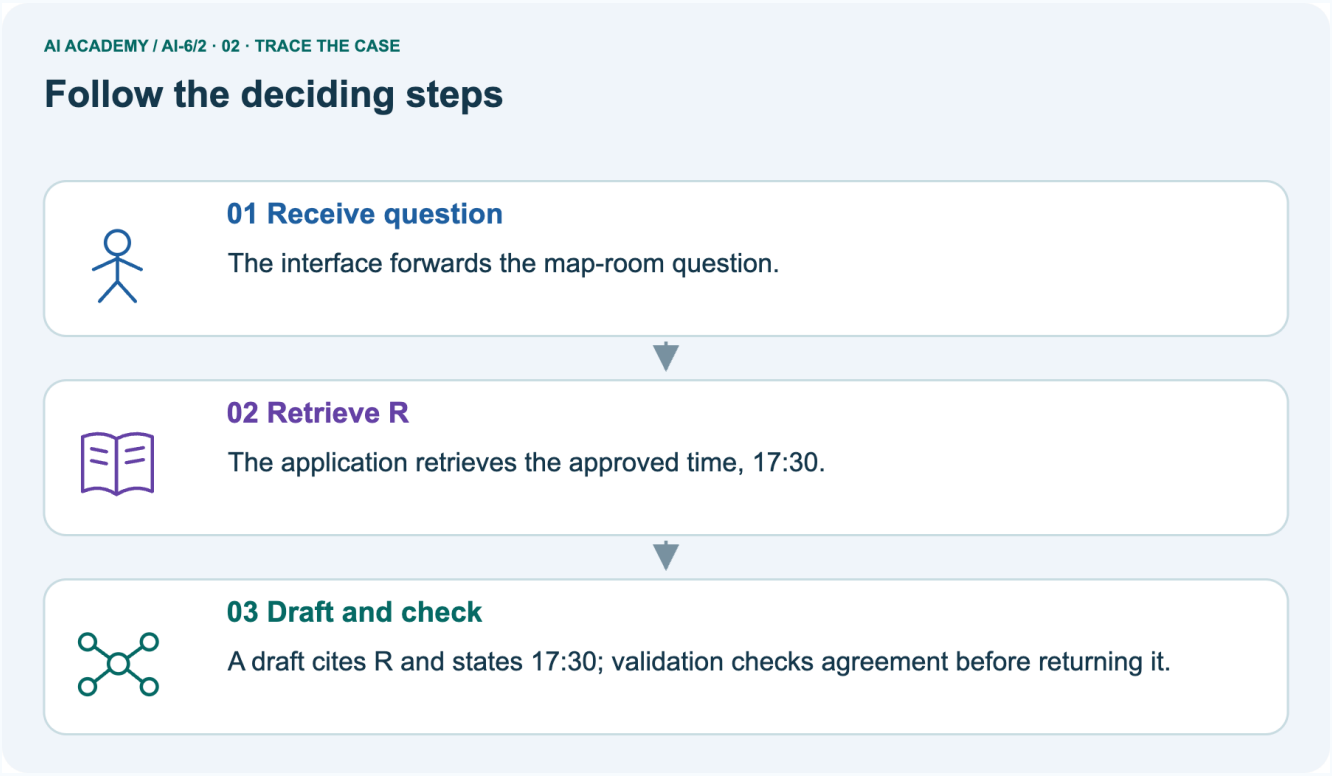


Follow the labelled picture with the matching explanation. Open the vector diagram

Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service. Supplied facts: Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

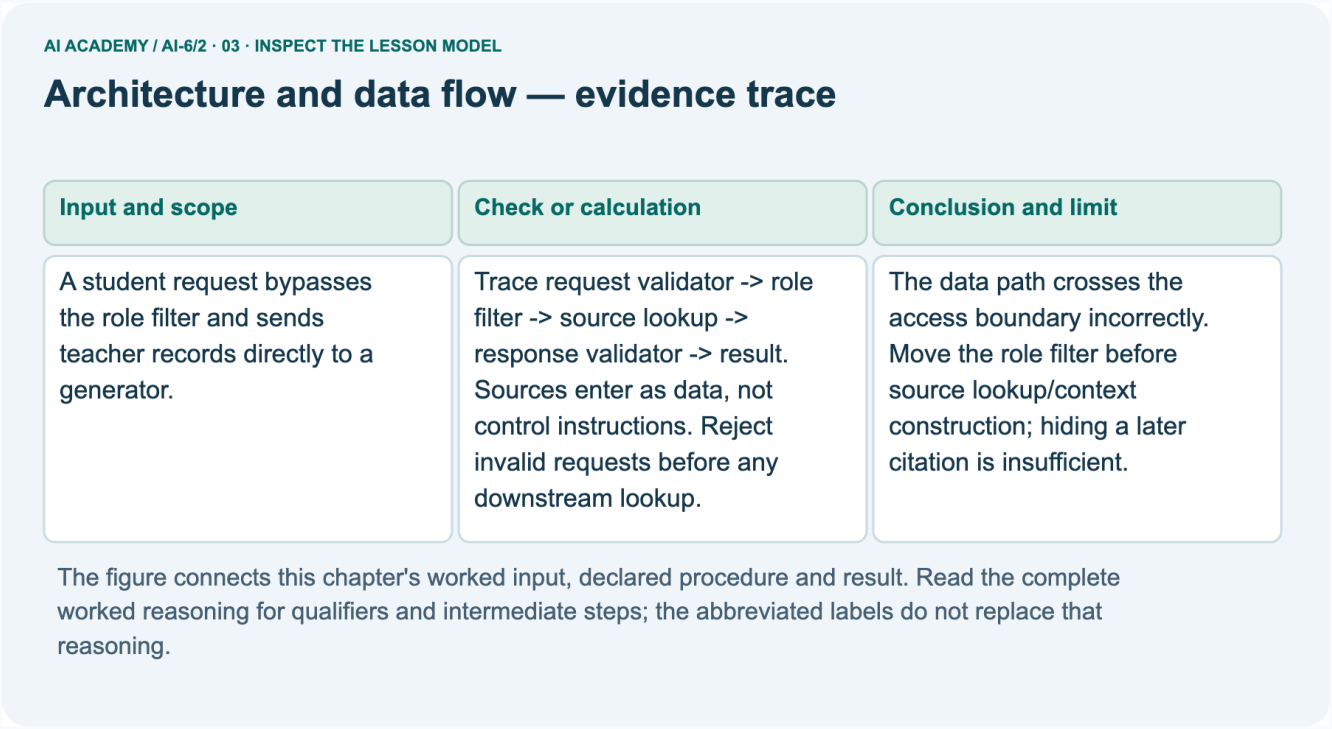


Follow the labelled picture with the matching explanation. Open the vector diagram

Receive question: The interface forwards the map-room question. → Retrieve R: The application retrieves the approved time, 17:30. → Draft and check: A draft cites R and states 17:30; validation checks agreement before returning it.

Point to each numbered stage and explain which recorded **input** it uses.

03. Architecture and data flow — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Architecture and **data flow** — **evidence** trace.

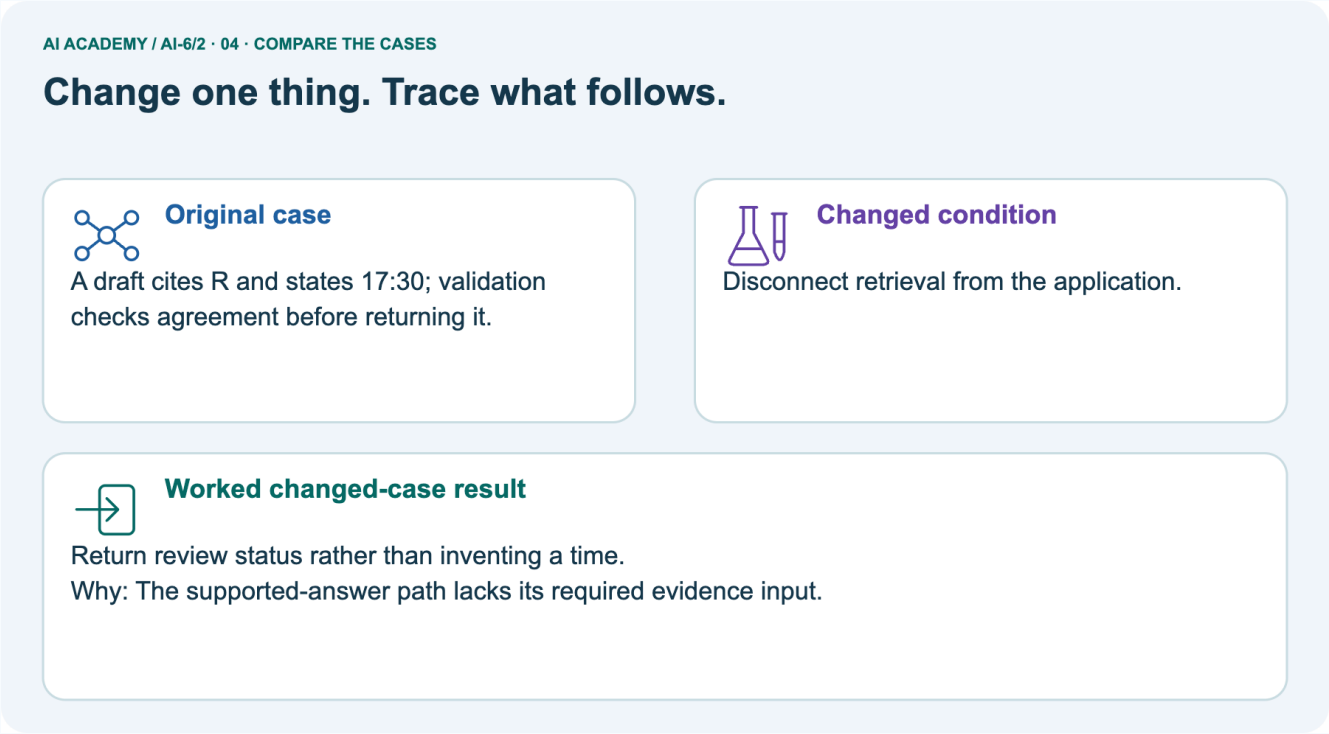
A student request bypasses the role filter and sends teacher records directly to a generator. | Trace request validator -> role filter -> source lookup -> response validator -> result.

Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup. | The data path crosses the access boundary incorrectly.

Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



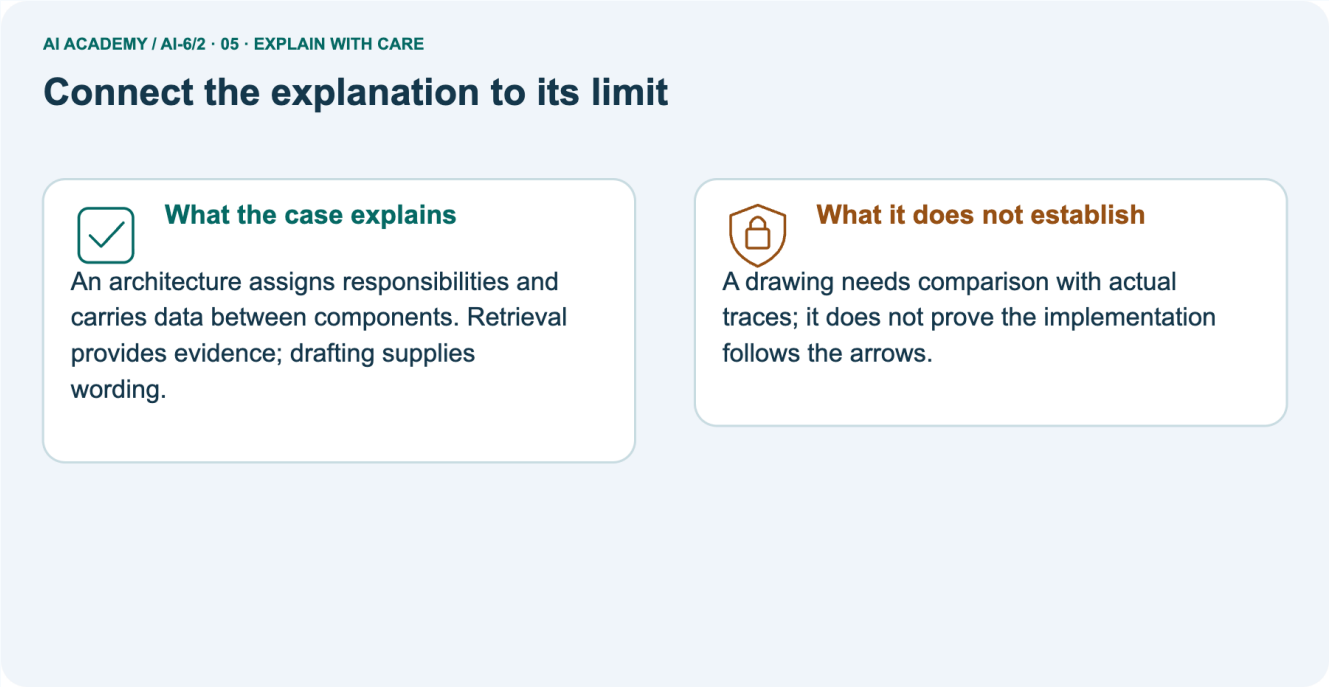
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A draft cites R and states 17:30; validation checks agreement before returning it. Changed condition: Disconnect retrieval from the application.

Expected result: Return review status rather than inventing a time. Reason: The supported-answer path lacks its required **evidence input**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



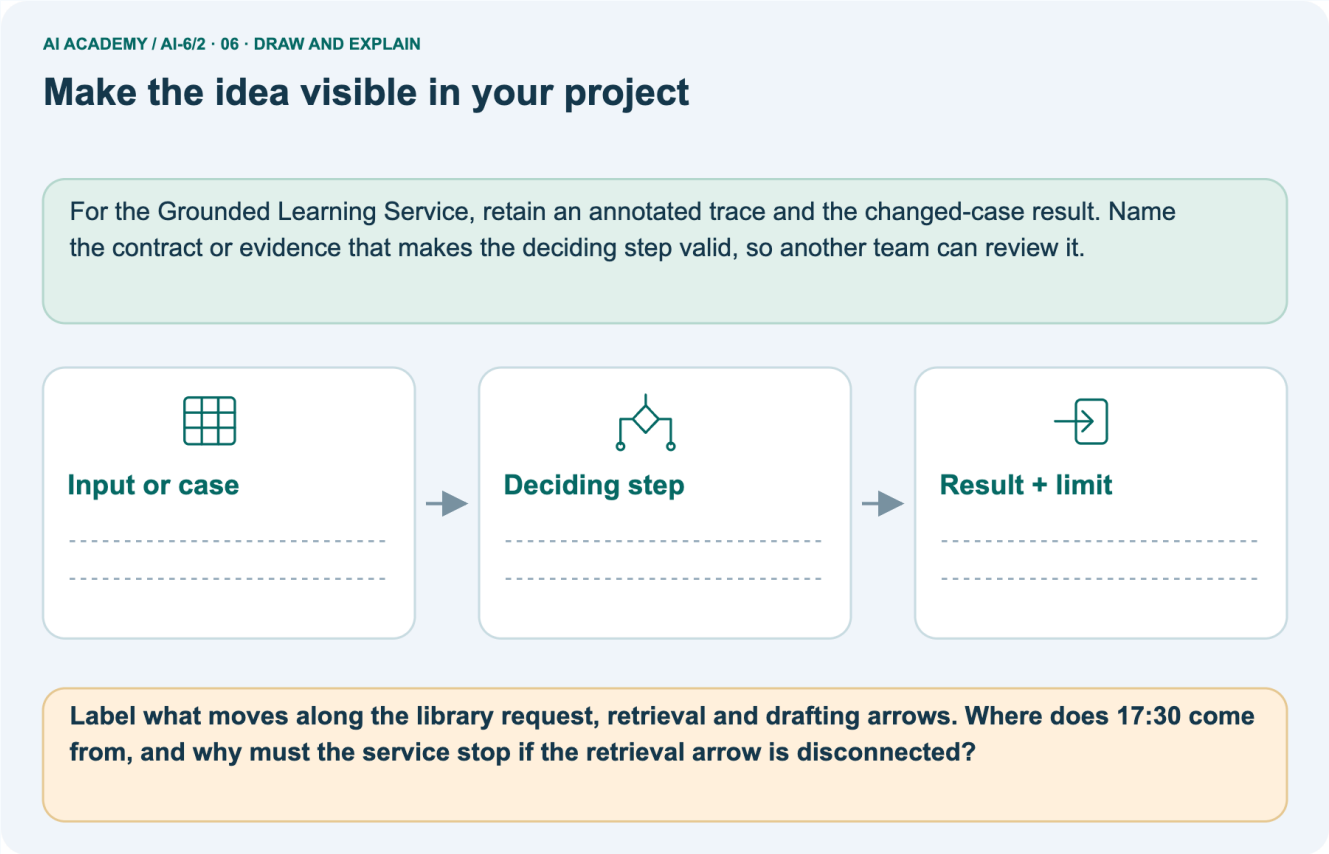
Follow the labelled picture with the matching explanation. Open the vector diagram

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

Limit: A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Architecture and data flow: the case

02. Follow the deciding steps

03. Architecture and data flow — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.